ALGEBRA

Upper × Upper = Upper

Lower × Lower = Lower

Upper × Lower = Unknown matrix

Diagonal × Upper = Upper

Diagonal × Lower = Lower

Diagonal × Diagonal = Diagonal

Lower Transpose = Upper

Diagonal Transpose = Diagonal

Upper inverse = Upper

Lower Inverse = Lower

Diagonal Inverse = Diagonal

Upper Transpose = Lower

$A^2 = A \times A$

$A^{-3} = (A^{-1})^3 = A^{-1} \times A^{-1} \times A^{-1}$

If $A = A^t$ => Symmetric Matrix

If $A = -A^t$ => Skew -Symmetric Matrix

For any $A_{(m\times n)}$ matrix A^tA , AA^t are symmetric matrices

(For the same matrix)

Symmetric × Symmetric ……. = Symmetric Matrix

Symmetric × Symmetric × Symmetric ……. = Symmetric Matrix

Skew × Skew × Skew × Skew … (Even times) = Symmetric Matrix

Skew × Skew × Skew … (Odd times) = Skew -Symmetric Matrix

AX = B, A is singular n × n matrix => infinity or no sol.

AX = B, A is non-singular n × n matrix => unique sol.

AX = 0, A is singular n × n matrix => infinity sol. (including trivial sol.)

AX = 0, A is non-singular n × n matrix => trivial sol.

AX = BX, A and B are non-singular n × n matrices => consistent

AX = BX, A and B are singular n × n matrices => consistent = homogenous

AX = BX, A is singular & B is non-singular n × n matrices => consistent

A = LU

AX = B => L(UX) = B => LZ = B , get Z => UX = Z , get X

U = upper matrix (elements above main diagonal) => $a_{ij} = 0 ; i > j$ ➡ $\begin{vmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{vmatrix}$

L = lower matrix (elements below main diagonal) => $a_{ij} = 0 ; i < j$ ➡ $\begin{vmatrix} 2 & 0 & 0 \\ 3 & 4 & 0 \\ 5 & 0 & 1 \end{vmatrix}$

$A_{2\times3} = \begin{matrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{matrix}$

$B_{3\times2} = \begin{matrix} j_{11} & j_{12} \\ j_{21} & j_{22} \\ j_{31} & j_{32} \end{matrix}$

$T_{2\times2} = A \times B = \begin{matrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{matrix} \times \begin{matrix} j_{11} & j_{12} \\ j_{21} & j_{22} \\ j_{31} & j_{32} \end{matrix}$

Entry in row 2 col 2 in T = ($a_{21} \times j_{12} + a_{22} \times j_{22} + a_{23} \times j_{32}$) and so on…

$I=E_K \ldots\ldots E_2E_1A$

$A^{-1}=E_K \ldots\ldots E_2E_1$

$A=E_1^{-1}E_2^{-1} \ldots\ldots\ldots E_K^{-1}$

$U=E_u \ldots\ldots E_2E_1A$

$A=(E_1^{-1}E_2^{-1} \ldots . E_u^{-1})U=LU$

$L=E_1^{-1}E_2^{-1} \ldots . E_u^{-1}$

MARO